Session Objectives

- Understand the important clinical aspects of a child suspected of having a GU problem
- Discuss the work up, treatment and management of a child suspected of having a GU problem
- Discuss various systemic disease states in children and implications to the renal system
- Discuss various structural problems in the GU system in a child and their clinical significance
- Understand enuresis in children and the various treatment modalities

Source: Course Syllabus

When to suspect kidney issues?

- Abnormal urine by history
- ***Abnormal urinalysis***
- Abnormal exam- peri-orbital swelling, generalized swelling, scrotal edema
- Hypertension
- Fever
- Flank pain

Hematuria

- Gross Hematuria?
- Trauma?
- Back pain?
- Fever?
- Menses?
- Family History?

- Work up?
  - Rpt UA
  - Urine Cx
  - Complete metabolic Panel (BUN/Cr)
  - Spot Ca/Creatine Ratio (<0.2nl)
  - 24hr urine for Ca/Cr
  - RUS
  - Other imaging:CT scan/ MRI
  - Nephrology consult
  - Urology consult

Treatment/ Workup

Likely **Minimal Change Dz**

- Corticosteroids

- When to Biopsy?
  - Recurrent relapsers
  - Steroid resistant
  - Nephritis (HTN)
- Dietary- NI protein, low salt

Nephrotic Syndrome- Watch out for…

Infections

- Fever and abdominal pain
  - **Peritonitis** (Strep Pneumoniae)
- Sepsis, pneumonia/ cellulitis

Hypercoagulable state

Post Streptococcal Glomerulonephritis (PSGN)

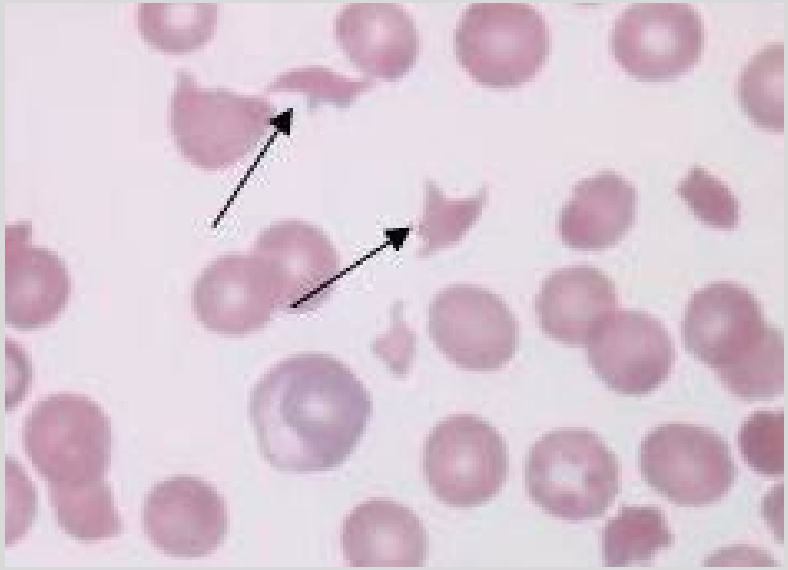
- Sudden onset of gross hematuria
- Edema
- Hypertension
- Oliguria
- Low Complement level
- Evidence of strep
  - ThCx/ Rapid strep
  - Antistreptolysin O titers (ASO)
  - Anti-DNAse B Antibody



Hemolytic Uremic Syndrome

- Hemolysis
  - Microangiopathic Anemia- Schistocytes
  - High Reticulocyte count
- Uremia
  - Oliguria
  - High BUN/Cr, proteinuria/hematuria/casts
- Petechiae/Purpura
  - Thrombocytopenia
- Bloody Diarrhea
  - Fever

- Neurological (less common)
  - Confusion
  - Irritable
  - Stroke
  - Relation with TTP



Hemolytic Uremic Syndrome

Infection

*** **Ecoli 0157-H7** – shigatoxin***

- Shigella
- StrepPneumonia
- Drug Toxicity

Treatment- supportive care if Ecoli (dialysis, IVF, transfusions)

- Antibiotics?...NO!
  - Potential to cause more damage by increasing toxin load

OUTBREAKS COMMON: Contaminated food and drink:

improperly cooked foods, contaminated drinking water, produce,
unpasteurized milk/ cheese

- 8 year old male presents to the office with abdominal pain. You notice him looking uncomfortable walking hunched over holding lower abdomen. Physical exam including Abdominal exam is benign….

You check his genitalia to complete the exam and….find it is his genital area that is hurting him



Diff Diagnosis of **Scrotal Pain**

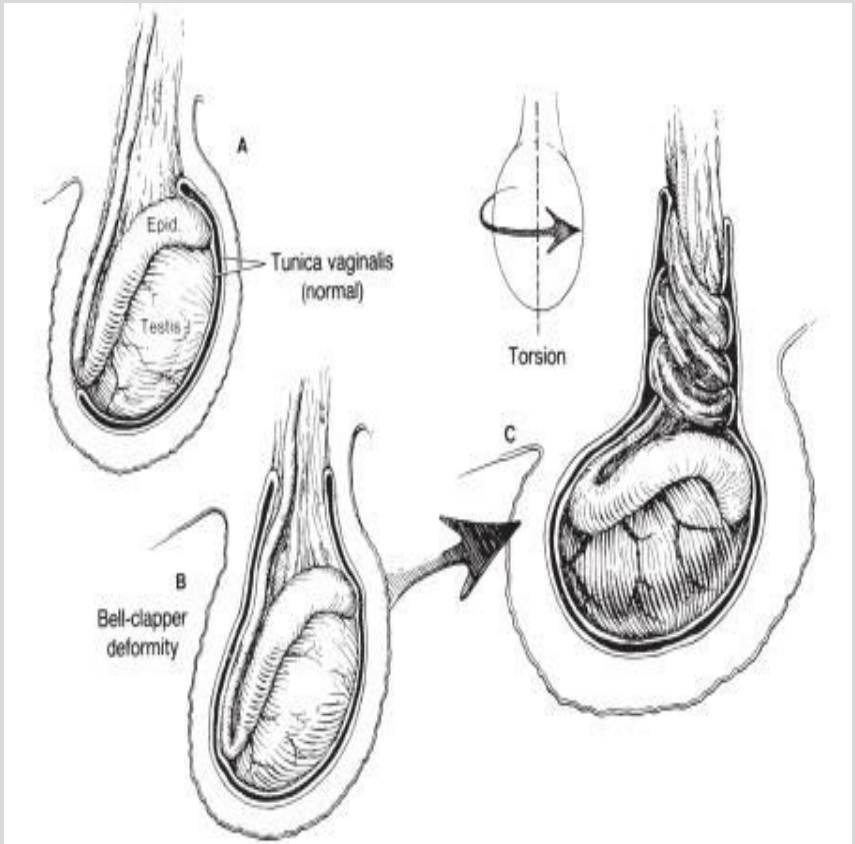
- Testicular Torsion
- Incarcerated Hernia
- Acute Epididymitis
- Orchitis
- Torsion of appendix testes
- Trauma
- Other…



Dx Testicular torsion- race against the clock!

Always check testicles in abdominal or groin pain!!!!!

- Swollen testicle
- High riding
- Only have 6-8hrs from onset of pain

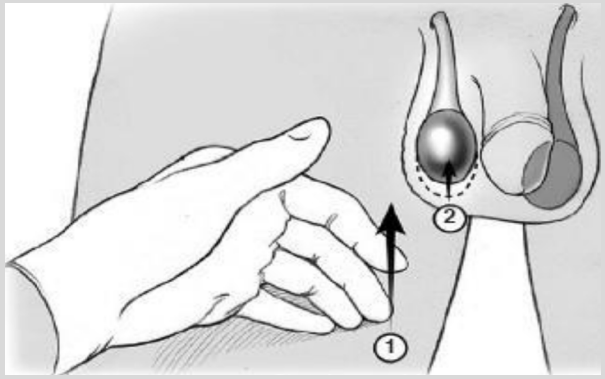


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Testicular Torsion- Crematerіc and Prehn



Figure 3. Cremasteric Reflex



The re ex is elicited by (1) stroking the ipsila eral inner high with a tongue depressor or gloved hand, resulting in (2) the elevation of he testicle throughcontrac on of the cremasteric muscle.
From: Davis JE. Chapter 31: Scrotal Pain, in Mahadevan Sli, *Garmel GM, eds. An Introduction to Clinical Emergency Medicine: Guide for Prac itioners in the Emergency Department. New York: Cambridge University Press; 2005:p466 . Reprinted with the permission of ca m bridge Universi ty Press.*

- **Absent cremasteric reflex**

Prehn Sign

- -ve Prehn's sign indicates no pain relief with lifting the affected testicle, which points towards testicular torsion which is a surgical emergency and must be relieved within 6 hours.
- Positive Prehn's sign indicates there is pain relief with lifting the affected testicle, which points towards epididymitis.

- **Negative/Absent Prehn sign (unreliable)**

Imperforate Hymen

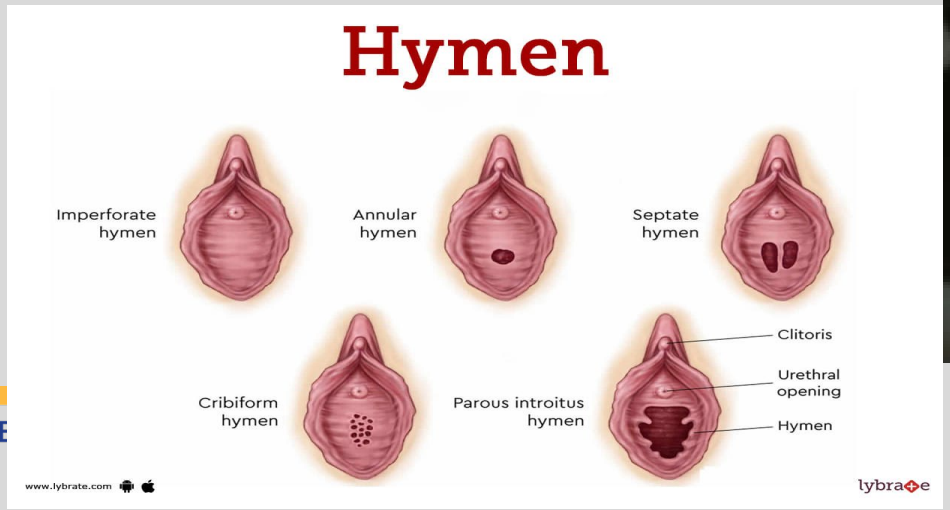
Often unrecognized till adolescence, sometimes recognized as a whitish bulge in neonates

Adolescent with recurrent monthly abdominal pain, back pain, urinary issues- can be severe over time and cause infertility issues, hydronephrosis/ kidney damage (mechanical obstruction of urinary tract)

Bulging hymen- often bluish from blood (hematocolpos)

Need surgical correction- Hymenectomy





- 8 year old boy is wetting the bed almost every night. He has always wet the bed and parents thought he would grow out of it but never has. Otherwise well. Parents are frustrated and child is embarrassed.

Primary Enuresis

- Never got full control of bladder
- 'Problem' after 6 years old
- Family Hx -similar in parent

Summary Slide 1

- Early morning puffy, swollen eyes
- HTN in children think kidneys
- Hematuria- send Ca/Creatine ratio
- Proteinuria send Protein/ Creatine ratio
- Post Strep GN- C3 low should return to nl by 6-8 weeks, if not look for other reason
- Non blanching *rash to buttocks*, joints, fever, abdominal pain- HSP
- Hemolytic Uremic Syndrome- hemolysis (anemia), uremia (dark or decreased urine, abnormal BUN/Cr), syndrome (fever, bloody stools, systemic complaints) –Ecoli 0157- Don't give antibiotics

Summary Slide 2

- Always check testicles in Abdominal pain/ Inguinal pain
- Testicular torsion- get Ultrasound with doppler

  (You only have 6-8 hrs from start to save the testicle)
- Blue dot sign- torsion of Appendix testes- Reassurance
- Abdominal/ pelvic pain in adolescent female- consider ovarian torsion in DiffDiag, if recurrent/monthly ab pain and no menses- think imperforate hymen
- Secondary enuresis- think psychosocial issue
- Primary enuresis- Bed wetting alarm, reward